

# Logic gates problems 7

| A | B | C | D | S |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | X |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | X |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | X |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | X |
| 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

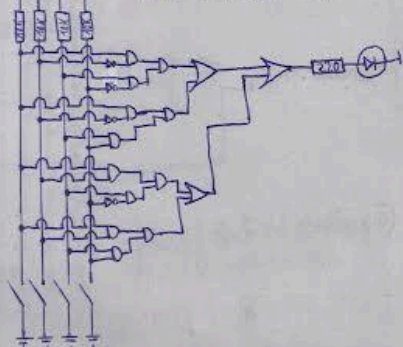
A = La presión interna de la cabina por debajo de valor atmosférico = 1

B = presencia de partículas contaminantes en el recinto = 1

C = puerta de acceso a vehículos bloqueada = 1

D = operario accionando pistola de pintura = 1

$$S = (A \cdot \bar{B} \cdot C \cdot \bar{D}) + (A \cdot \bar{B} \cdot C \cdot D) + (A \cdot B \cdot C \cdot \bar{D}) + (A \cdot B \cdot C \cdot D)$$



| AB | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 | 0  | 0  | 0  | 0  |
| 01 | X  | X  | X  | X  |
| 11 | 0  | 0  | 1  | 1  |
| 10 | 0  | 0  | 1  | 1  |

$$S = A \cdot C$$

